



Semester Two Examination, 2021

Question/Answer booklet

MATHEMATICS APPLICATIONS UNITS 1&2

Section One: Calculator-free

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free**35% (52 Marks)**

This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1**(5 marks)**

A fruit picker is paid 60 cents for each kilogram of cherries they pick, and a box of cherries contains 9 kg of the fruit.

(a) Determine, in dollars, how much a picker is paid to pick one box of cherries. (2 marks)

(b) Determine how much a picker earns in a day when they pick 40 boxes of cherries. (1 mark)

(c) A picker worked for four days and earned a total of \$660. Calculate how many kilograms of cherries they picked in this time. (2 marks)

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Question 2

(6 marks)

- (a) The surface area S of an object is given by $S = 2(wh + lh + wl)$.
Calculate the value of S when $h = 8$ cm, $l = 4$ cm and $w = 0.5$ cm. (2 marks)

- (b) The quantity Δ is calculated using the formula $\Delta = \sqrt{b^2 - 4ac}$.
Calculate the value of Δ when $a = 2$, $b = 7$ and $c = 3$. (2 marks)

- (c) Given that $a^2 - b^2 = (a + b)(a - b)$, evaluate $42^2 - 32^2$. (2 marks)

Question 3**(6 marks)**

The weights in kilograms of the 10 members of basketball squad P are shown below. The standard deviation of these weights is 6.3 kg.

79, 81, 73, 67, 62, 78, 73, 68, 77, 82.

- (a) Determine the mean weight of the members of squad P. **(2 marks)**

Basketball squad Q also has 10 members. The mean and standard deviation of the weights of the members of this squad are 77 and 7.9 kg respectively.

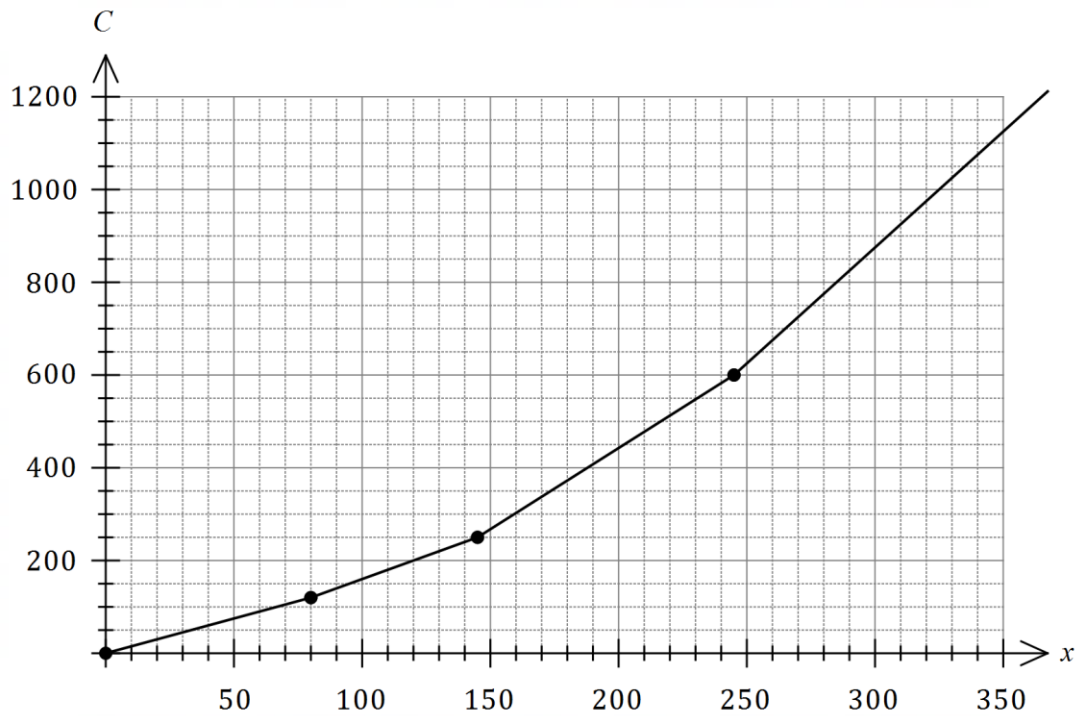
- (b) Write a statement, with reasoning, that compares the variation in weights of the members of the two squads. **(2 marks)**

- (c) Write a statement, with reasoning, that compares the weights of the members of the two squads. **(2 marks)**

Question 4

(6 marks)

The annual cost of water, $\$C$, varies with x , the total number of kilolitres of water used by a household in a year, as shown in the graph below.



- (a) Determine the cost of water for a household that used 275 kL in a year. (1 mark)
- (b) Calculate the slope of the line between (145, 250) and (245, 600) and interpret, in context, this slope. (3 marks)
- (c) Describe how the slope of the piecewise graph changes as the yearly household water use increases and suggest, in context, a reason for this change. (2 marks)

Question 5

(7 marks)

Let $\mathbf{P} = \begin{bmatrix} 1 & 2 \\ -3 & -2 \end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix} 2 & 1 \\ -4 & 3 \end{bmatrix}$.

(a) Express $\mathbf{P} + 2\mathbf{Q}$ as a single matrix.

(2 marks)

(b) Calculate the matrix $\mathbf{P} \times \mathbf{Q}$.

(2 marks)

(c) Determine the value of a , the value of b and the value of c given the matrix equation

$$a \begin{bmatrix} 2 & 5 & -3 \\ 3 & 1 & b \end{bmatrix} + \begin{bmatrix} 1 & -1 & 4 \\ 3 & 5 & 3 \end{bmatrix} = \begin{bmatrix} -1 & c & 7 \\ 0 & 4 & 1 \end{bmatrix}.$$

(3 marks)

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Question 6**(7 marks)**

- (a) The area of a trapezium can be calculated using the formula shown below.

$$A = \frac{h}{2}(a + b)$$

Determine the value of a when $A = 75$, $h = 5$ and $b = 17$.

(3 marks)

- (b) To hire a scooter from company A, a customer is charged a fee of \$8 plus \$5 per hour, but to hire one from company B, they are charged a fee of \$10.50 plus \$3 per hour.

Ben hires four scooters from company B and Aida hires three scooters from company A. If they both hire the scooters for the same length of time, x hours, and their bills are the same, use the above information to write an equation and hence determine the value of x .

(4 marks)

Question 7

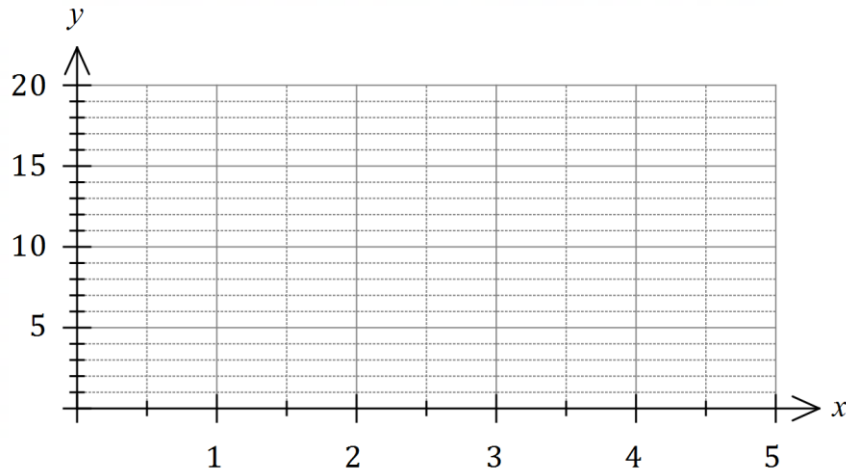
(7 marks)

Water is pumped into a storage tank. The table below shows the depth of water, y cm, in the tank after the pump has been running for x minutes.

x	1	2	3	4
y	7	11	15	19

- (a) Plot these points on the axes below.

(1 mark)



- (b) What feature of the graph in part (a) suggests a linear formula relates x and y ?

(1 mark)

- (c) The formula $y = ax + b$ relates the depth of water to the time. Determine the value of the constant a and the value of the constant b .

(2 marks)

- (d) Determine the depth of water in the tank after 15 minutes.

(1 mark)

- (e) Determine the time required for the depth of water in the tank to reach 91 cm.

(2 marks)

Question 8**(8 marks)**

A farmer grew two varieties of potato, Pontiac and Snow Gem. The time for each crop to mature is normally distributed with the times shown in the following table.

Variety	Pontiac	Snow Gem
Mean time for crop to mature (days)	95	104
Standard deviation of time for crop to mature (days)	5	4

- (a) Use deviations from the mean (standard scores) to explain which of the following would be more unusual: a crop of Pontiac maturing in 101 days; or a crop of Snow Gem maturing in 101 days. **(3 marks)**

- (b) Use the 68%, 95%, 99.7% rule to determine the approximate probability that

(i) a crop of Pontiac takes less than 95 days to mature. **(1 mark)**

(ii) a crop of Snow Gem takes between 92 and 116 days to mature. **(2 marks)**

(iii) a crop of Pontiac matures in more than 105 days. **(2 marks)**

Supplementary page

Question number: _____

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